

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc. I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VI

Roll No.: T018	Name: Sachin Ramesh Mehta
Class: TYIT	Batch: 1
Date of Assignment: 30/07/2026	Date/Time of Submission: 30/07/2026

- a. **Create a simple website for a College Information Portal and convert it into a Progressive Web Application.**

Code:

manifest.json:

```
{
  "name": "College information portal",
  "short_name": "college portal",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

offline.html:

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <meta charset="utf-8" />
5    <title>Offline</title>
6  </head>
7  <body>
8    <h2>Please check your internet connection </h2>
9  </body>
10 </html>
```

service-worker.js:

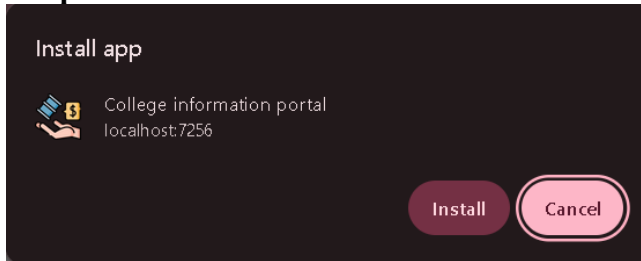
```
const CACHE = "v1";

self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/offline.html"
      ]);
    })
  );
});

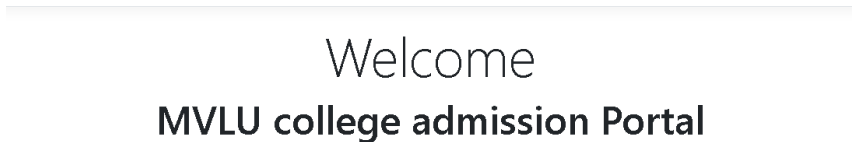
console.log("Service Worker Installed");

self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch(() => caches.match("/offline.html"))
  );
});
```

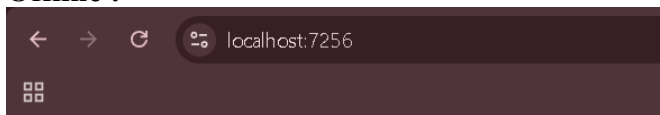
Output:



Online:



Offline :



Please check your internet connection

- b. Create a PWA that displays a custom offline page when the user is not connected to the internet.

Code:

manifest.json:-

```
{
  "name": " HTML TO PWA",
  "short_name": "HTP",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "/icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "/icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

service-worker.js

```
self.addEventListener("install", e => {
  console.log("Service worker installed");
});

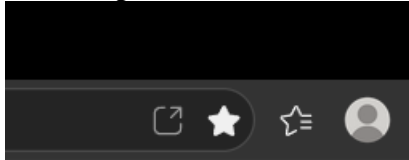
self.addEventListener("fetch", e => {
  e.respondWith(fetch(e.request));
});
```

Layout.cshtml:-

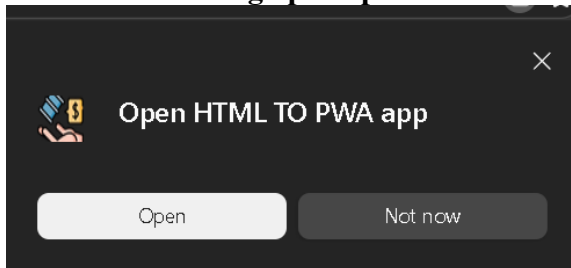
```
<Link rel="manifest" href="/manifest.json" />
<script>
    if('serviceWorker' in navigator) {
        navigator.serviceWorker.register('/service-worker.js');
    }
</script>
```

Output:

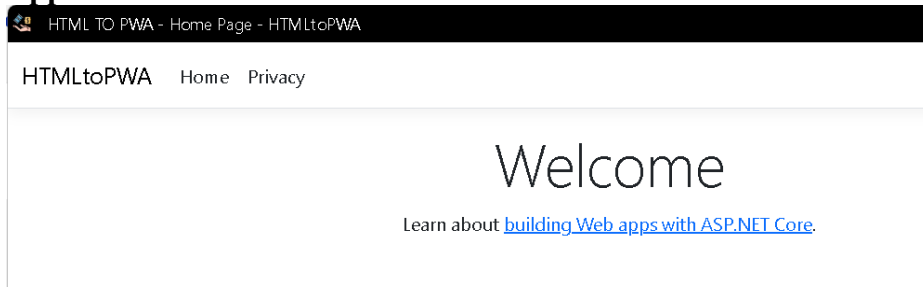
Install option available:



After downloading open option:



Application in another tab:



In application tab click on three dot this will appear:

